nudgely↗

Team Nudgely · AI-GNITION 2026
Problem Statement 1 · Built on MantaGO (Eslitec)

Message the right customers, say the right thing, prove it worked.

Nudgely turns the purchase and chat data a business already holds into personal MantaGO messages. It sends them only to customers a message will actually bring back, and measures the lift against a no-message group.

+57% visits
Email vs no email · 64k shoppers
Hillstrom randomised test. Purchases about 1.9x

+7% visits
SMS vs no SMS · 687k grocery customers
Lenta randomised test. 10.3% → 11.0%

53% vs 30% random
Extra visits captured by our top 30%
About 1.8x. Noisy: 90% interval 31–89%

PROBLEM AND TARGET USERS

Small retailers and F&B businesses send the same *"We miss you, 10% off"* to everyone and can't tell whether it brought anyone back. Discounts are wasted on customers who would have returned anyway, messages ignore what customers have told the shop, and nobody measures the effect.

Target users: owners and marketers of small businesses that talk to customers on WhatsApp, Instagram, Messenger or LINE through MantaGO. Demo business: **Kampung Fresh**, a fictional neighbourhood grocer in Tampines.

SOLUTION AND CORE WORKFLOW

1. **Smart Send (who).** An uplift model predicts how much more likely each customer is to return *because of* a message, picks the top customers within budget, and skips those who'd return anyway or won't be moved.
2. **Chat Memory (what).** An LLM reads past chats and extracts tags (restock request, delivery complaint, upcoming party). Every tag must quote the customer's own words.
3. **Personal message.** One short draft per customer: discount capped at 10%, apology first for complaints, opt-out line. The owner previews each one.

   > Hi Aisha! Planning your son's birthday party? We've got party snack packs ready for you! 🎉 Enjoy 10% off your next order too. Reply STOP to opt out

4. **Send via MantaGO.** The Open API delivers each message (WhatsApp) and can store the tags on the customer's profile.
5. **Prove it.** A random 10% of chosen customers is held out. Retention lift = return rate of messaged minus held-out customers.

EVIDENCE

FINDING	RESULT
Email vs no email Hillstrom · 64k online shoppers	Visits 10.6% → 16.7% (+57%); purchases 0.57% → 1.07% (about 1.9x). Best email: +72% visits, 2.2x purchases
SMS vs no SMS Lenta · 687k grocery customers	Visits 10.3% → 11.0% (+7%)
Our top 30% vs random 30% Lenta test set · 103k unseen customers	Captures 53% of all extra visits vs 30% (about 1.8x). 90% bootstrap interval 31–89%

Expected impact: more returns per message, fewer wasted discounts on customers who'd come back anyway, fewer paid WhatsApp messages, and a monthly retention-lift number the owner can trust.

TOOLS AND TECHNOLOGY

MantaGO Open API · scikit-uplift · scikit-learn HistGradientBoosting · Google Gemini API · Python · Streamlit · Streamlit Community Cloud

Model trained offline on Lenta. LLM output is JSON, auto-checked against hard rules and cached. The app never calls the LLM on page load.

RESPONSIBLE AI

- **Consent.** Live sends go only to our own opted-in test phones. Opt-outs are blocked automatically. Every message ends "Reply STOP to opt out".
- **Privacy.** Chats are scrubbed of names and phone numbers before the LLM reads them. No API keys in GitHub.
- **Fairness.** Targeting uses behaviour only: gender, age and children are removed from the model.
- **Human in control.** The owner previews every message and must confirm before sending. Rehearsal mode is the default.
- **Singapore.** A real rollout adds recorded consent and DNC Register checks under the PDPA, following the Model AI Governance Framework's human-in-the-loop approach.

LIMITATIONS AND NEXT STEPS

- **Demo data:** names and chats are fictional. Results come from public overseas datasets (Russian grocery SMS, US online email), not a live Singapore business.
- **Noisy estimate:** the true lift is small, so the 53% has a wide interval. A real pilot needs 2–4 weeks with a holdout group.
- **Needs one past test:** uplift models learn from a message vs no-message campaign. A new business starts with one random test month.
- **Next:** pilot with one MantaGO business in Singapore, read reply events automatically, and retrain monthly on the business's own results.